Capital University
Faculty of Engineering
Computer & Systems Engineering Department

FitGuard

AI-Powered Fitness and Nutrition Coach

Project Assignment 02

Mobile Computing Lab — Spring 2026

Course Information
Course: Mobile Computing Lab
Instructor: Eng. Rania El-Sayed
Semester: Spring 2026 | 4th Year — Computer Engineering

Team Members	
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Problem Statements and Proposed Solutions

FitGuard addresses three core problems faced by fitness enthusiasts who lack professional guidance, proper form correction, and nutritional awareness.

01 Poor Exercise Form Leading to Injuries

Problem

Many gym-goers, especially beginners, perform exercises with incorrect posture and form. Without real-time guidance, this leads to muscle strain, joint injuries, and long-term physical damage that prevents continued training.

Proposed Solution

FitGuard uses AI-powered computer vision through the smartphone camera to analyze the user's body posture in real time during workouts. It detects incorrect angles in the back, knees, and hips, then provides instant visual feedback to correct form before injury occurs.

02 Lack of Personalized Fitness Guidance

Problem

Personal trainers are expensive, unavailable 24/7, and inaccessible to most people. Generic workout apps provide identical plans for all users regardless of individual fitness level, physical condition, or personal goals.

Proposed Solution

FitGuard's AI Coach analyzes each user's profile — including weight, height, fitness goal, activity level, and recovery score — to generate a fully personalized daily workout plan. The system adapts recommendations based on progress and adjusts intensity when recovery is insufficient.

03 Poor Nutritional Awareness and Diet Tracking

Problem

Most people who exercise regularly do not track their nutrition properly. They either under-eat, over-eat, or consume incorrect macro ratios, which prevents them from achieving their fitness goals despite consistent training.

Proposed Solution

FitGuard includes an intelligent Nutrition module that tracks daily calorie intake and macro breakdown (Carbohydrates, Protein, Fat). Users can log meals manually or use the Scan Food feature to identify food via the camera. The app then suggests meals aligned with the user's daily caloric goal and dietary preferences.

Design System

Color Palette — utils/colors.js

The following colors are defined in utils/colors.js and applied consistently throughout all screens.

#7C3AED **Primary Purple** Buttons, active states, highlights	#3B0764 **Purple Dark** Header backgrounds, deep accents	#EDE9FE **Purple Light** Backgrounds, badges, soft fills	#0D0D1A **Background Dark** Main application background	#1E2235 **Card Surface** Cards, input fields, navigation bar
#F59E0B **Gold Accent** Statistics, streaks, achievement icons	#10B981 **Success Green** Correct form, completed goals	#EF4444 **Error Red** Incorrect form, warnings, alerts	#00C2FF **AI Blue** AI Coach card gradient accent	#6B7280 **Muted Gray** Subtitles, placeholders, captions

Typography — assets/fonts/fonts.js

Four font roles are defined in assets/fonts/fonts.js to maintain a consistent visual hierarchy across the application.

Title	Header	Regular	Bold
Bebas Neue	**Montserrat Bold**	**Montserrat Regular**	**Montserrat SemiBold**
32 – 48px	**20 – 24px**	**14 – 16px**	**14 – 18px**
Application name, splash screen, large display headings	Screen titles, section headers, card titles	Body text, descriptions, form labels	Button labels, statistics, highlighted values

Custom Buttons — components/CustomButton.js

All custom button components are defined in components/CustomButton.js in the main project root.

Component	Style	Used In
PrimaryButton	Filled purple, white bold text, rounded	Login, Sign Up, Start Plan, Chat with Coach
SecondaryButton	Transparent background, white border	Continue as Guest, Cancel actions
SocialButton	Dark surface, icon + white text, rounded	Continue with Google, Continue with Apple
ScanFoodButton	Purple pill shape with QR icon	Scan Food in Nutrition screen
IconActionButton	Square dark card with icon and label	Quick Actions: Workout, Scan Food, Log Water, Weigh In

Wireframe — Figma

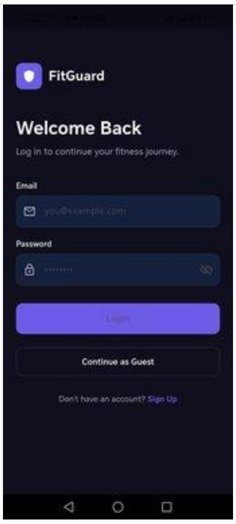
Interactive Wireframe — Figma

6 Screens: Login | Sign Up | Home | Workouts | Nutrition | Nutrition Details

https://www.figma.com/design/GB1O5D2VpuwZdfZD8Po7Ec/FitGuard_Wireframe?node-id=0-1&t=nOzl3RRweVANerAT-0

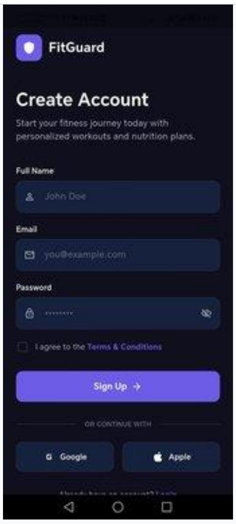
Application Screens

The following screens represent the initial implementation of the FitGuard application, organized in the screens/ folder of the project root.



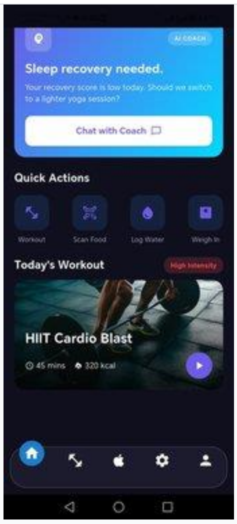
01 Login Screen

Entry point for returning users. Email and password fields with show/hide toggle. Options to login, continue as guest, or navigate to sign up.



02 Sign Up Screen

New user registration with full name, email, and password. Includes terms agreement checkbox and social sign-in options (Google and Apple).



03 Home Screen

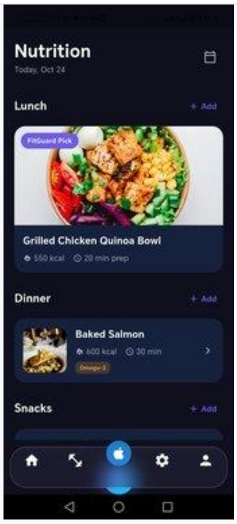
Main dashboard displaying activity summary (steps, calories, active time), AI Coach recommendations, quick action shortcuts, and today's workout card.

04 Workouts Screen

Workout library with filter tabs (All, At Home, Gym, No Equipment). Shows AI-recommended plans, featured workout cards with intensity labels and duration.

05 Nutrition Screen

Daily nutrition tracker showing caloric progress, macro rings (Carbs, Protein, Fat), and meal sections (Breakfast, Lunch, Dinner, Snacks) with food cards.



06 Nutrition Details

Extended nutrition view showing detailed meal cards with food images, calorie counts, preparation time, dietary tags, and the Scan Food feature.